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Are public transport ınformation systems equally effective? Insights from local and migrant passengers

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Abstract. This study investigates the differences in public transport usage patterns and perceptions of passenger information systems between migrant and non-migrant populations in Türkiye. Based on data collected from 513 participants through face-to-face surveys conducted in the cities of Sakarya and Istanbul, the research adopts a descriptive approach to explore demographic characteristics, transport mode preferences, and satisfaction with digital and on-site information services. The analysis reveals that migrants tend to be younger, more highly educated, and are more likely to use high-capacity transport modes such as metro systems. Interestingly, migrants also report higher satisfaction with information systems, particularly with mobile applications and language accessibility—an unexpected finding considering the generally limited multilingual support in Türkiye's public transport infrastructure. This may reflect positive experiences in cities like Istanbul, where such systems are more developed, as well as comparative perceptions shaped by migrants' countries of origin. Common challenges shared by both groups include long waiting times and concerns about vehicle safety. The study emphasizes the importance of inclusive, technology-supported transport systems and suggests that future research should expand to other cities—both metropolitan and less developed—in order to assess how infrastructure disparities influence public transport accessibility and user satisfaction among diverse urban populations.

Keywords: Public transport; Passenger ınformation systems; Immigirants; Accessibility

1. Introduction

Urban transportation systems play a fundamental role in ensuring social inclusion, economic vitality, and quality of life in modern cities. Efficient and accessible public transport (PT) services not only facilitate daily mobility but also contribute to broader societal goals such as environmental sustainability and social equity. In an increasingly diverse urban landscape, the inclusiveness of transportation systems—encompassing infrastructure, vehicles, information systems, and supporting digital applications—has become a central concern for both local authorities and policymakers.

For transportation systems to serve all segments of society effectively, they must address the needs of diverse user groups, including both native residents and migrants. Public transport inclusiveness is not limited to physical accessibility; it also encompasses the effectiveness of passenger information systems, the clarity of real-time announcements, the user-friendliness of route planning applications, and the availability of multilingual services (Sime & Fox, 2014). For migrants, who may face linguistic, cultural, and informational barriers, the ability to easily access and interpret public transportation information is a critical factor influencing their mobility patterns and integration into urban life (Li et al., 2024; Welsch et al., 2016).

Despite the recognized importance of inclusive transportation design, relatively limited research has focused on comparative analyses of migrants' and non-migrants' experiences with public transport, particularly in Türkiye's urban settings. Understanding the differences and commonalities between these groups can provide valuable insights into how transportation services—and especially digital passenger information systems—can be improved to better serve an increasingly heterogeneous population.

This study aims to contribute to this emerging field by conducting a descriptive analysis of public transport usage patterns among migrants and non-migrants. Based on survey data, the research examines how demographic factors correlate with transport habits, the perceived effectiveness of infrastructure and digital information tools, and the role of passenger information systems in facilitating urban mobility. Through this lens, the study seeks to highlight critical areas for enhancing the inclusiveness of public transportation services in Türkiye.

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2. Literature review

Public transport systems play a crucial role in fostering social inclusion by enhancing accessibility, expanding infrastructure, and integrating digital passenger information tools. Academic research highlights these components as essential for creating equitable transportation networks that cater to the needs of diverse populations, including marginalized groups.

One foundational aspect of social inclusion in public transport is the accessibility of such systems. Saghapour et al. (2016) elaborate on how inadequate access to public transportation can hinder individuals' abilities to engage with various services and activities, leading to social isolation and exacerbating socio-economic disadvantages (Saghapour et al., 2016). This aligns with Yang et al., who advocate for comprehensive accessibility indices that consider various demographic factors to ensure that public transport serves all urban residents effectively (Yang et al., 2019). These infrastructural advancements are critical for building a connected urban environment that facilitates movement for all citizens.

The increasing integration of digital technology into public transport systems has significant implications for social inclusion. Digital tools can disseminate real-time information about services, enhancing the user experience and encouraging public transit usage (Davidsson et al., 2016). Digital maps and applications have been cited as effective means to promote green transportation, encouraging sustainable commuting behaviors among users, particularly in urban settings where congestion and emissions are concerns (Zhang et al., 2024). Furthermore, digital systems can streamline payments and expand access to transport services, making them more appealing to a broader audience, as evidenced by findings from Sener et al., which indicate that prior positive experiences with public transit foster ongoing use intentions (Sener et al., 2019). Research indicates that an integrated and well-structured public transport system enhances social cohesion among diverse communities.

Migrants face numerous barriers when using urban public transport systems, which significantly impacts their ability to integrate and access vital services. These barriers are often categorized into three primary areas: language difficulties, access to information, and navigation challenges. Insufficient proficiency in the local language can impede migrants' understanding of transit systems, leading to confusion and hesitation in using public transport. According to Nteliou et al., language proficiency directly influences the ability of migrants to navigate public facilities, including transport systems, exacerbating issues of exclusion and integration (Nteliou et al., 2021).This lack of understanding can deter them from accessing information necessary for effective public transport use.

Access to relevant information regarding transport options and services is another critical barrier faced by migrants. As indicated by Waitt et al., many migrants lack adequate information about public transport schedules, routes, and payment systems, which can hinder their mobility (Waitt et al., 2016). This situation is often worse in rural areas where public transport services are limited or non-existent (Pilling & Estes, 2016). In metropolitan areas, digital information tools have the potential to alleviate some informational barriers. However, the efficacy of these tools depends on their accessibility and usability for migrants. Studies indicate that many public transportation systems fail to provide multilingual and culturally relevant information, which can further marginalize non-native speakers and create disparities in service usage (Gruer et al., 2021).

Navigating public transport systems poses unique challenges for migrants, particularly those who may be unfamiliar with urban environments. Navigational challenges arise from inadequate signage, the complexity of transit maps, and a lack of user-friendly tools that cater to migrants. As highlighted by Welsch et al., migrants are less likely to own personal vehicles, making them more reliant on buses and trains; however, the unfamiliarity with the transport landscape can deter them from using these services (Welsch et al., 2016). Moreover, the experience of feeling lost or insecure in an unfamiliar environment can deter migrants from fully utilizing public transport systems. This navigation issue is compounded by the fact that many urban transport systems still operate with outdated or insufficient technology that does not accommodate the diverse needs of their users (Welsch et al., 2016).

The effectiveness of passenger information systems and mobile applications in improving public transport accessibility and user experience for diverse populations has garnered considerable attention. Recent studies highlight the various dimensions of how these digital tools facilitate better mobility and enhance the overall user experience across different demographics, including migrants, seniors, and individuals with disabilities. Real-time passenger information systems significantly enhance the user experience by reducing uncertainty and making travel more predictable (Corsar et al., 2018). Similarly, mobile applications have emerged as powerful tools that enhance public transport accessibility. The participatory design approach in creating mobile apps, as discussed by Vollenwyder et al., ensures that the needs of users with disabilities are met by incorporating their feedback into app development (Vollenwyder et al., 2020). This fosters user engagement and enhances the functionality of the digital tools, making transportation more accessible to all demographics. Furthermore, Setiabudi and Tjahyana (2015) demonstrate that hybrid mobile applications can improve public transport user experiences by addressing individual preferences and facilitating better planning through intuitive interfaces.

On the other hand, as migrants become more established within their new environments, their dependency on public transport tends to decrease, indicating a shift towards car ownership as financial stability increases (Blumenberg & Evans, 2010). This transition underscores the dynamic relationship between transportation

availability, socio-economic status, and migrant integration processes. Cities that emphasize developing equitable public transport systems are better equipped to handle the influx of diverse populations and foster a sense of community among residents (Sime & Fox, 2014).

The implications of inclusive transport systems extend beyond mere mobility; they also encompass broader social integration. Inclusive transport facilitates access to essential services, such as healthcare and education, thereby enhancing migrants' quality of life and social connectivity (Li et al., 2024). Hence, it is evident that building inclusive public transport frameworks can play a transformative role in enhancing the social fabric of urban environments as they yield benefits not only for migrants but also for non-migrants by creating more interconnected communities.

3. Methodology

This study adopts a descriptive research design to examine public transport usage patterns and perceptions among migrants and non-migrants in Türkiye. The primary objective is to explore how demographic characteristics relate to transport habits and the use of passenger information systems, without applying inferential statistical modeling.

3.1. Data collection

Data were collected through a structured survey instrument administered via face-to-face interviews. The survey was conducted in two major metropolitan cities, Sakarya and Istanbul, in November and December 2024. These locations were selected due to their diverse populations and significant levels of immigration, which provide a rich context for analyzing public transport accessibility and usage patterns among different demographic groups.

A total of 513 valid responses were obtained. Participants were approached in public areas such as transport hubs, community centers, and marketplaces to ensure a diverse representation of public transport users. Participation was voluntary, and respondents were assured of anonymity and confidentiality. The questionnaire included items covering participants' demographic profiles, transportation usage frequency, types of public transport utilized, satisfaction with passenger information systems, and the use of mobile applications related to public transportation.

Survey items consisted of both categorical variables (e.g., gender, age, education level, employment status) and Likert-scale items (ranging from "Strongly Disagree" to "Strongly Agree") that assessed perceptions of accessibility, information availability, and overall service quality.

3.2. Sample characteristics

The sample includes individuals representing a broad range of socio-demographic backgrounds. Respondents were classified into two main groups based on self-reported citizenship status:

• *Non-migrants:* Native Turkish citizens born in Türkiye.

• *Migrants:* Individuals who were either born outside of Türkiye or had migrated within the last 10 years, regardless of their citizenship status.

This classification allows for a comparative descriptive analysis between migrants and non-migrants regarding their transport experiences and informational needs.

3.3. Data analysis

The analysis was exclusively descriptive. Frequencies, percentages, and cross-tabulations were employed to summarize the data across the key demographic and usage variables. Initially, general demographic distributions were examined to outline the profile of respondents. Subsequently, descriptive comparisons were made between migrant and non-migrant groups in terms of transport usage frequency, satisfaction with transport services, and experiences with passenger information systems.

4. Findings and results

The findings of this study provide a descriptive overview of public transport usage patterns, perceptions of passenger information systems, and demographic characteristics among migrants and non-migrants residing in Sakarya and Istanbul in Türkiye. Based on data collected through face-to-face surveys, the results highlight key differences and similarities between these two groups. The analysis focuses on how variables such as gender, age, education level, and household income relate to public transport behavior and access to information systems. By examining these patterns, the study aims to shed light on the inclusiveness of urban mobility services and the extent to which current public transport infrastructures meet the needs of diverse populations. This section first presents the demographic profile of the respondents, followed by a comparative evaluation of public transport usage and satisfaction across migrant and non-migrant groups. Table 1 summarizes the demographic distribution of migrants and non-migrants in the sample, categorized by gender, age, education level, and household income. The table reports both the number and percentage of respondents within each group, providing a basis for understanding the population diversity among public transport users.

Table 1. Demographic characteristics of survey respondents

Category	Migrant	Migrant Percentage	Non-Migrant	Non-Migrant Percentage
Gender				
Male	105	29,91	246	70,09
Female	64	40,25	95	59,75
DNPS	1	33,33	2	66,67
Age				
< 18	13	44,83	16	55,17
18-24	79	50,00	79	50,00
25-34	38	28,15	97	71,85
35-44	27	26,21	76	73,79
45-54	8	14,81	46	85,19
55-64	3	20,00	12	80,00
65+	2	10,53	17	89,47
Education				
Middle school or less	9	8,11	102	91,89
High school	23	17,69	107	82,31
College or above	138	50,74	134	49,26
Household income				
<17,000 TRY	21	70,00	9	30,00
17,000–29,000 TRY	52	52,00	48	48,00
30,000–44,000 TRY	38	26,39	106	73,61
45,000–59,000 TRY	32	25,00	96	75,00
60,000–79,000 TRY	15	19,74	61	80,26
80,000–99,000 TRY	1	5,26	18	94,74
100,000+ TRY	11	68,75	5	31,25

*DNPS: Do not prefer to say

The following Fig. 1 illustrates vehicle ownership status among migrants and non-migrants. A significant disparity is observed between the two groups. While 170 non-migrants reported owning a vehicle, only 33 migrants indicated the same. Conversely, 137 migrants reported not owning a vehicle, compared to 173 non-migrants. This suggests a notably lower rate of vehicle ownership among migrants, potentially reflecting economic disparities, differences in mobility needs, or access to private transportation. Such insights are critical for informing inclusive transportation and mobility policies targeting vulnerable populations.

On the other hand, Fig. 2 presents the distribution of public transport mode preferences among migrant and non-migrant respondents. Among non-migrants, the bus is clearly the most preferred mode, with 252 selections, followed by 224 for minibus, 65 for heavy rail/metro, 42 for tram. In comparison, migrants show a more balanced distribution across multiple modes. While the minibus (99) and bus (98) are the most frequently selected modes by migrants, the heavy rail/metro system also receives notable attention with 30 selections. This indicates that migrants make significant use of both formal and semi-formal systems and tend to diversify their public transport usage more than non-migrants, possibly due to their residential dispersion across urban peripheries and high-density corridors.

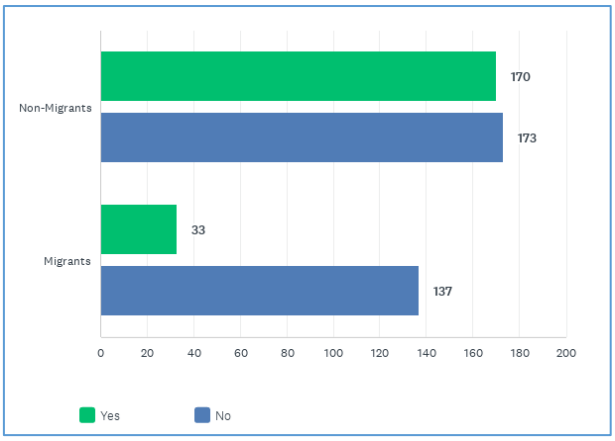

Fig. 1. Vehicle ownership by migration status

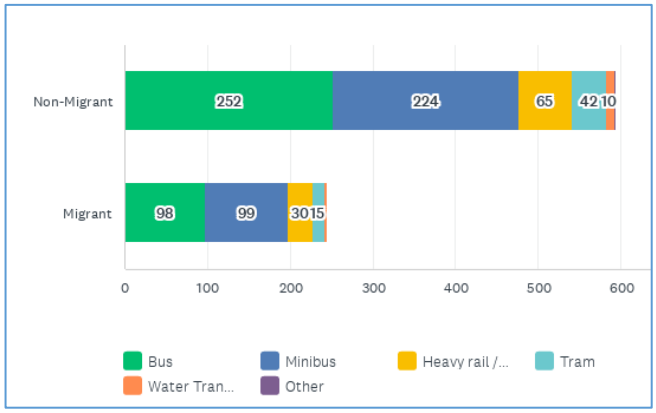

Fig. 2. PT mode/type preference

Fig. 3 compares the attitudes of non-migrant and migrant respondents toward public transport (PT) and associated information systems. Each statement was evaluated using a three-point Likert scale (Agree, Neutral, Disagree), allowing for a visual contrast between the two groups. Overall, both groups exhibit relatively positive perceptions; however, migrant respondents consistently demonstrate slightly higher levels of agreement across most items, indicating a more favorable evaluation of the transport services and digital tools available to them.

Regarding general public transport usage, both groups show high levels of agreement with the statements *"I prefer to use PT in my daily life"* and *"Using PT is cost-effective for me"*. These findings highlight the essential role that public transport plays in the daily routines of both populations. However, the slightly higher agreement among migrants suggests that they may be more reliant on public transport for economic and practical reasons, likely due to lower rates of private vehicle ownership and a greater need for affordable mobility.

Satisfaction with public transport operations—particularly in terms of routes and schedules—also appears to be slightly higher among migrants. Migrant respondents more frequently agree that *"The routes and schedules of PT meet my needs"* and that *"I can easily access up-to-date information about PT schedules."* This may be attributed to their more active use of mobile applications or their greater sensitivity to transport reliability as newcomers navigating unfamiliar environments.

Interestingly, a notable and somewhat unexpected difference between the two groups emerges in their evaluation of passenger information systems. Migrants report higher satisfaction with onboard announcements and the availability of language options compared to non-migrants. This finding is surprising given that passenger information systems in Türkiye are often perceived as inconsistent, with limited multilingual support and uneven technological integration, especially outside major urban centers. However, it is possible that some migrant respondents reside in Istanbul, where relatively advanced and accessible transport infrastructure—including digital displays, audio announcements, and mobile app integration—is more widely available.

Moreover, the higher levels of satisfaction among migrants may also be influenced by their comparative perspective: when contrasted with the public transport environments in their countries of origin, Türkiye's infrastructure—particularly in metropolitan contexts—may represent a notable improvement in terms of clarity, modernity, and accessibility. This contrast could enhance their perception of effectiveness, even if the system still presents gaps from a local policy or universal design standpoint. Therefore, this result underscores not only the

importance of localized infrastructure investment, but also how relative expectations and past experiences shape perceived quality among diverse user groups.

In relation to mobile technologies, both groups express strong approval of mobile applications as tools for planning and navigating public transport. Statements such as *"I regularly use PT mobile applications on my smartphone"* and *"I can quickly access PT schedules and route information"* received high agreement from both groups, with slightly greater enthusiasm from migrants. This reinforces the critical role of digital tools in enhancing the public transport experience, especially for users who may depend on real-time information to reduce uncertainty and plan their journeys efficiently.

Despite these generally positive evaluations, certain areas reveal shared challenges. Both groups identify *long waiting times during travel* as a persistent issue. Additionally, non-migrants express more concern regarding the *safety of PT vehicles*, with higher disagreement levels in this item. These findings suggest that while information systems and mobile applications are functioning effectively, there are still operational aspects of the public transport system—particularly reliability and safety—that require attention for all users.

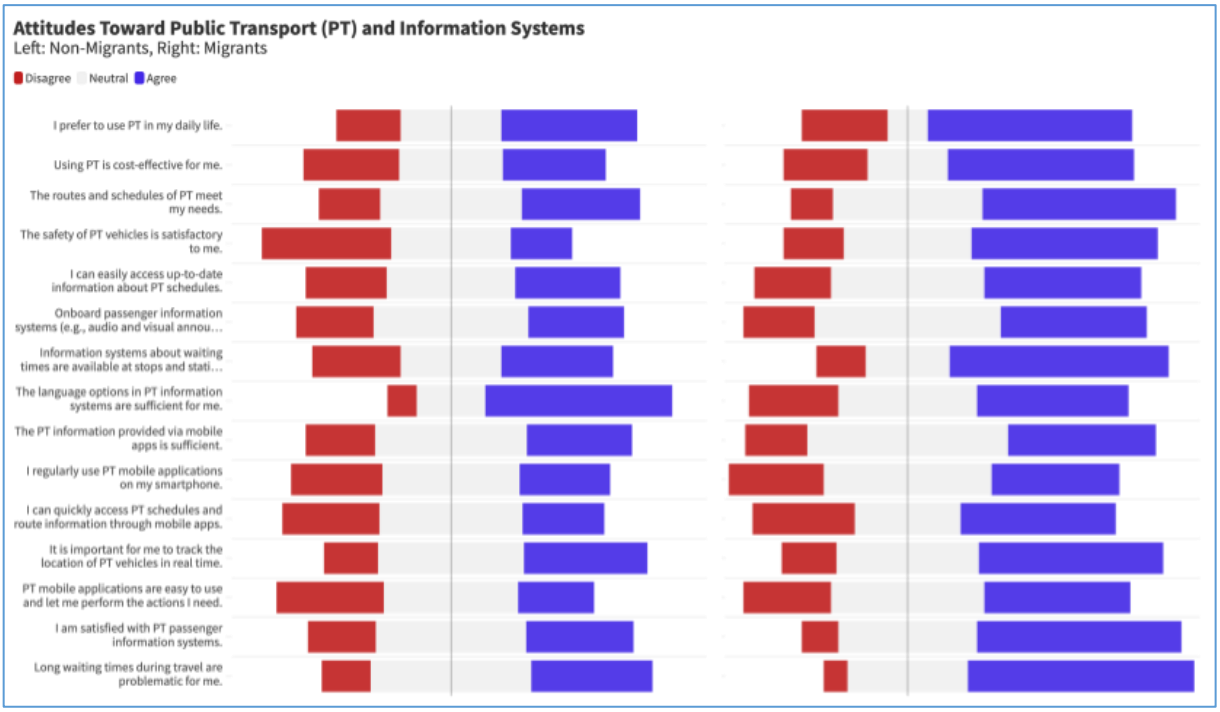

Fig. 3. Comparison of attitudes toward public transport (PT) and passenger information systems among non-migrant and migrant respondents.

5. Conclusion and discussions

This study provided a descriptive analysis of public transport usage patterns and perceptions of passenger information systems among migrant and non-migrant populations in Türkiye. Drawing on data collected through face-to-face surveys conducted in Sakarya and Istanbul, the findings offer valuable insights into how different user groups experience and evaluate urban mobility services.

The demographic analysis revealed that migrants tend to be younger, more likely to have higher educational attainment, and more concentrated in both the lowest and highest income brackets. These findings suggest a heterogeneous migrant population, consisting of both economically vulnerable individuals and highly mobile professionals or students. In contrast, non-migrants were generally spread in various age and income categories, reflecting a more stable, long-settled urban population.

Public transport usage patterns showed that both groups rely heavily on buses and minibuses; however, migrants reported greater use of metro systems, suggesting that they are more engaged with high-capacity and multimodal networks, particularly in metropolitan areas like Istanbul. Migrants also demonstrated slightly higher satisfaction levels with public transport routes, schedule information, and affordability, indicating a relatively positive experience despite broader system-level limitations in Türkiye.

A particularly notable finding was the migrants' more favorable evaluation of passenger information systems, especially in terms of onboard announcements, language accessibility, and mobile applications. This result was somewhat unexpected, given that Türkiye's information systems are often criticized for inconsistency and lack of multilingual support outside of major cities. However, this positive perception may stem from both the higher-

quality infrastructure available in cities like Istanbul and a comparative advantage when contrasted with the public transport conditions in migrants' countries of origin. These experiences highlight the role of digital tools and inclusive communication strategies in improving accessibility and reducing uncertainty, particularly for newer or more vulnerable urban residents.

Despite these positive trends, both migrant and non-migrant groups expressed dissatisfaction with waiting times and, to a lesser extent, concerns about vehicle safety. These shared pain points emphasize the ongoing need to improve not only digital and informational infrastructures but also the operational reliability of public transport services.

Future research could build on these findings by expanding the geographical scope beyond Istanbul and Sakarya. While this study offers valuable insights from two urban centers with relatively developed transport infrastructures, it remains important to examine whether similar patterns and perceptions emerge in other parts of Türkiye. Comparative studies that include other metropolitan cities such as Ankara, Izmir, or Bursa, as well as smaller urban centers with less advanced public transport infrastructure, would help determine the extent to which infrastructure quality influences migrant and non-migrant experiences. Moreover, exploring cities with limited multimodal integration or reduced availability of digital information systems could reveal whether satisfaction and accessibility gaps widen under less favorable transit conditions. Such efforts would contribute to a more comprehensive understanding of transport equity and informational inclusion across different urban contexts in Türkiye.

In conclusion, the study underscores the importance of maintaining and expanding inclusive, multimodal, and technology-enhanced transport systems that meet the needs of a diverse urban population. Policymakers should pay particular attention to the mobility challenges of migrants—not only through equitable route and fare planning, but also by ensuring that information systems are accessible, multilingual, and responsive to varying levels of digital literacy. Doing so will support more inclusive, efficient, and resilient public transport networks in Türkiye's rapidly urbanizing contexts.

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